

# COOL-AIR

HCI PROJECT 2 PRESENTATION



# PROBLEMS

## PROBLEMS

**1**

### **Limited Understanding of Air Quality Risks**

Residents, especially elderly and non-tech users, don't understand AQI data or know when outdoor conditions are dangerous.

**2**

### **Health Risks from Sudden Air Quality Changes**

People with asthma or chronic illnesses struggle to respond quickly when the air worsens, since existing alerts aren't personalized or timely enough.

**3**

### **Emotional Stress & Lack of Support During Smoke Events**

Limited knowledge on how to connect with volunteers or community aid.



# POV STATEMENTS

## POV 1

Elderly Resident needs a way to receive clear, spoken air-quality warnings because they don't use mobile apps and can't interpret AQI numbers.

## POV 2

Asthma Patient needs a way to receive personalized health alerts and reminders to take medication because bad air days can trigger symptoms suddenly.

## POV 3

A resident living near active wildfires needs a way to track changing air quality and get immediate evacuation or indoor safety instructions, because conditions can deteriorate quickly and official updates often arrive late.





**Mr. and Mrs. Lee: Elderly Residents  
Living Near Polluted Area**

# PERSONAS

## PERSONAS

**Age:** 74

**Occupation:** Retired schoolteachers

**Location:** Tai Po, Hong Kong (close to forested hills)

**Tech Comfort:** Low — uses a basic smartphone mainly for calls

**Languages:** Cantonese only

**Background:** Mrs. Lee lives alone in a small apartment without air conditioning. She often keeps her windows open without realizing how bad the air is. She gets confused by air quality terms like “PM2.5” and rarely checks news updates on time.

**Goals:**

Understand easily when air quality is unsafe

Receive spoken reminders and safety tips

Feel reassured and cared for during smoke alerts





**Jason: Asthma Patient and  
Office Worker**

# PERSONAS

## PERSONAS

**Age:** 28

**Occupation:** Marketing Executive

**Location:** Central

**Tech Comfort:** High — uses mobile apps regularly

**Languages:** English, Cantonese

**Background:** Jason has mild asthma and tracks his health closely. He uses air quality apps but finds them too technical. When wildfire smoke moves suddenly across Hong Kong, he sometimes only realizes too late when symptoms flare up.

**Goals:**

Get early warnings about worsening air

Receive personalized reminders for medication and mask use

Understand health risks in plain language





**Brian: The Concerned Entrepreneur**

# PERSONAS

## PERSONAS

**Age:** 42

**Occupation:** Owner of a small company in Northern California

**Location:** Lives near San Jose, recently affected by wildfire smoke

**Tech Comfort Level:** High – uses digital tools for both business and home management

**Background:** David runs a delivery and logistics company that relies heavily on outdoor work. When wildfires hit his area last summer, smoke and poor air quality forced him to temporarily close operations. His family had to stay indoors for days, and he became increasingly aware of how little information he had about local air quality and safety guidelines.

**Goals:**

Wants a trusted platform that alerts him early about changing air quality.

Needs guidance for business to work or transport goods safely.

Seeks community updates to understand how nearby residents are coping.



# **SELECTED IMPACTED COMMUNITY**

**People who are not at immediate risk of evacuation from forest fires, but are not in a position to leave their homes because of respiratory concerns.**



# ‘HOW MIGHT WE’ QUESTIONS

## **POV 1 → HMW**

How might we help elderly residents understand air quality risks through clear, spoken, and easy-to-follow guidance without relying on apps or technical data?

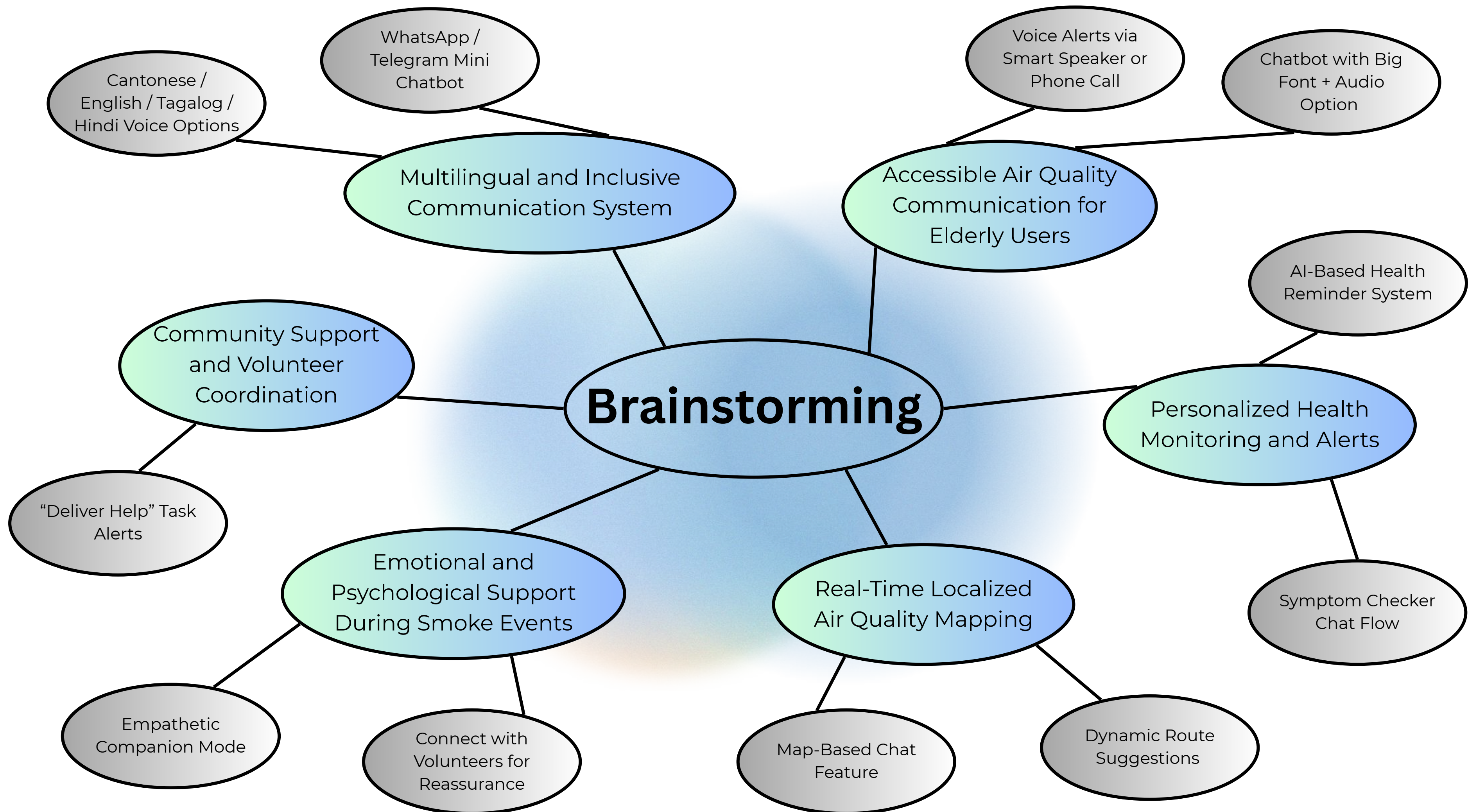
## **POV 2 → HMW**

How might we design a system that monitors local air quality and provides personalized, health-focused alerts and care reminders to people with respiratory conditions during wildfire smoke events?

## **POV 3 → HMW**

How might we create a system that delivers hyperlocal alerts about smoke movement and safety actions to residents living near wildfire zones?









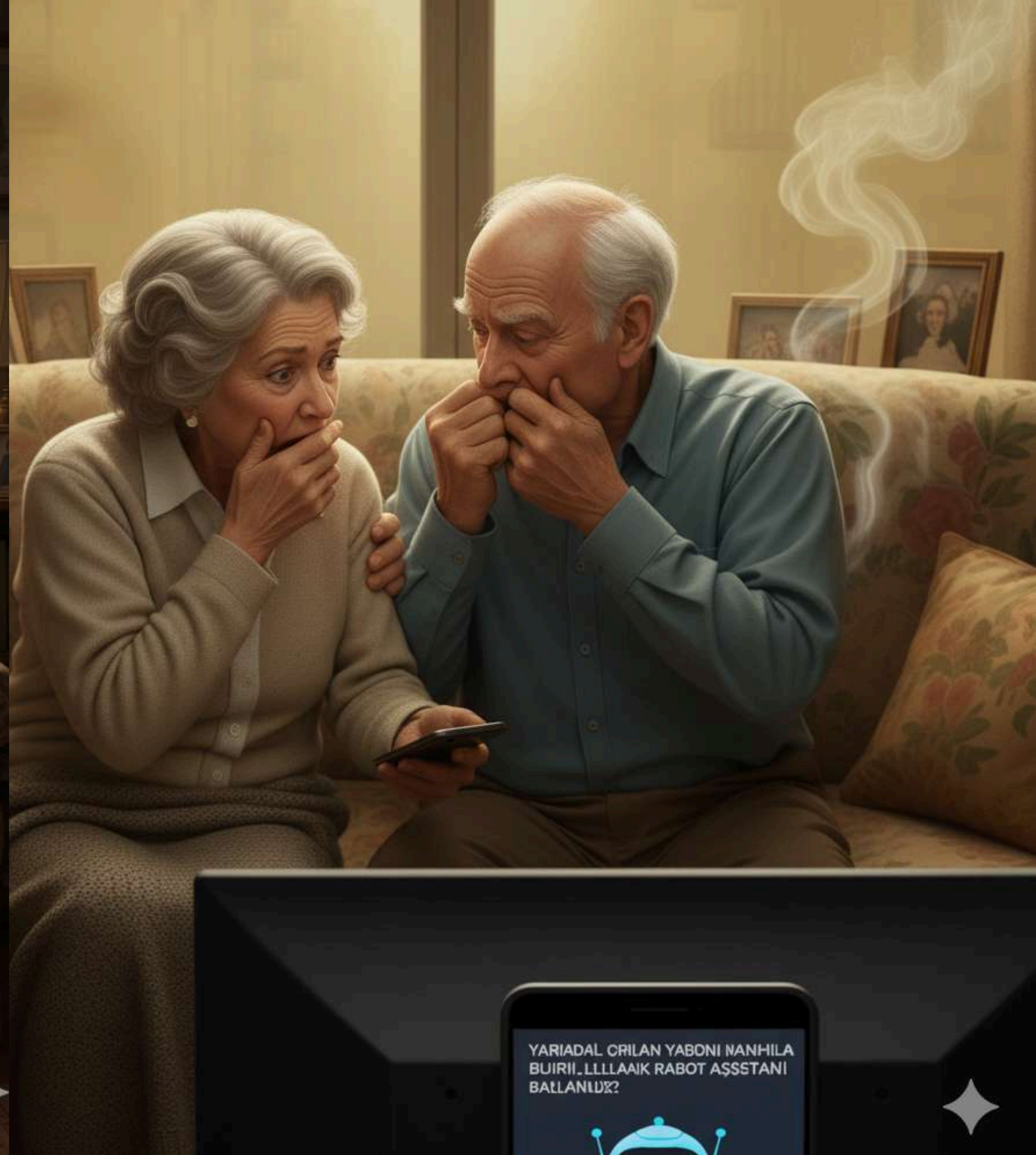
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**New Product**

# PRODUCT

# GOMS

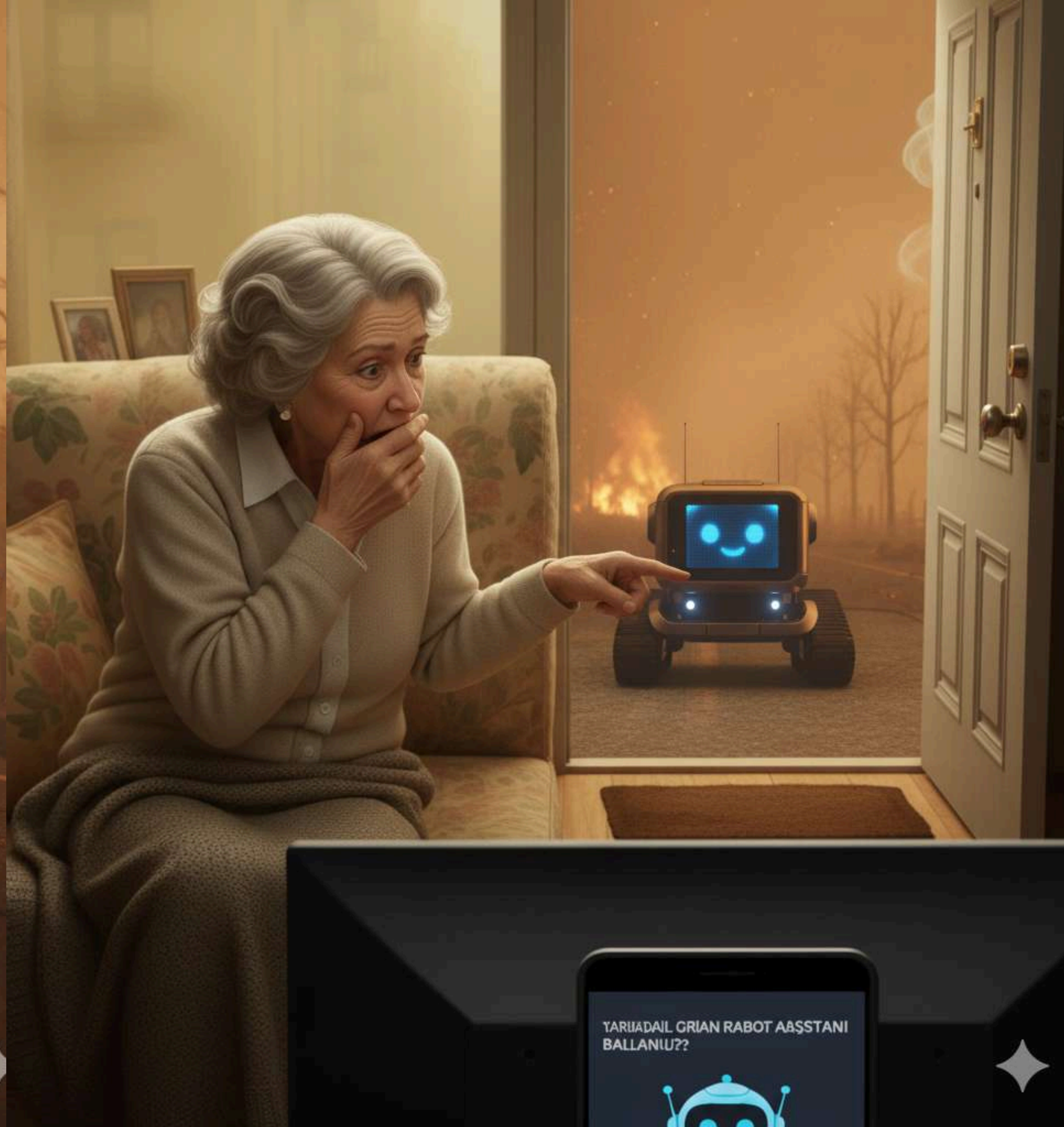




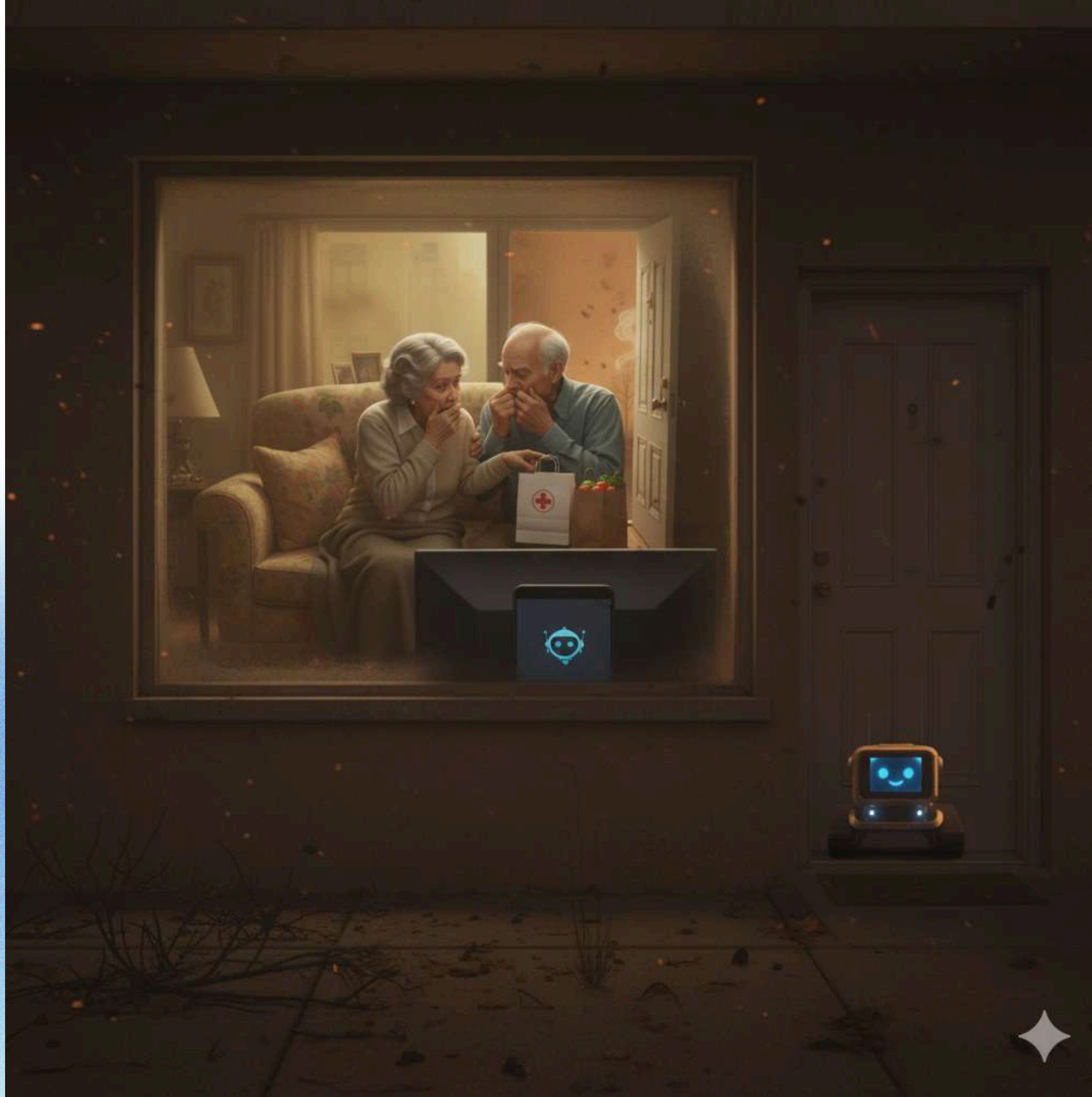
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**1**

STEP 1  
**2023**

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**2**

STEP 2  
**2024**

**3**

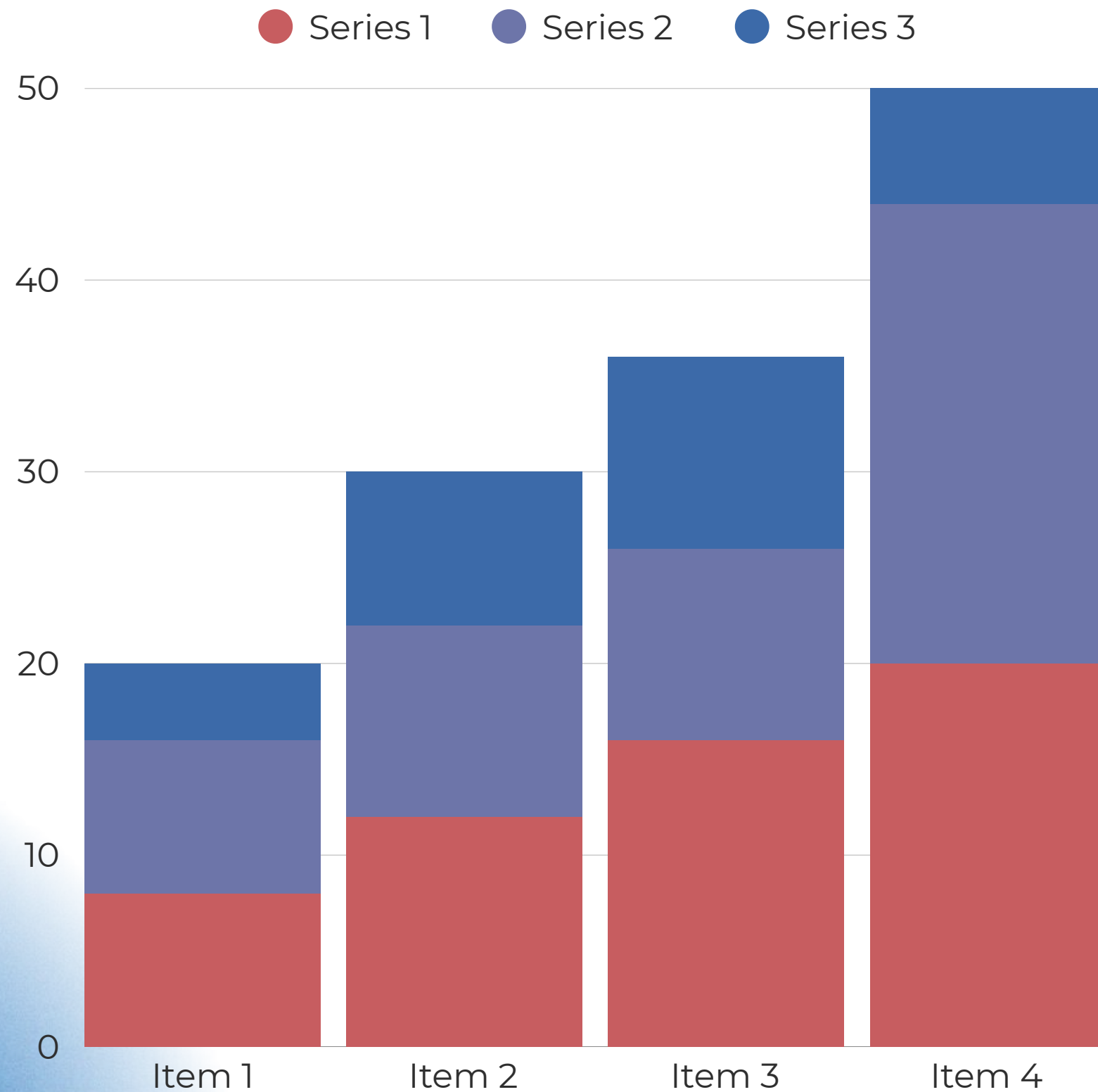
STEP 3  
**2025**

# **VIDEO SHOWCASE**



# FEEDBACK

## USER TESTING



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# QUESTIONS?

PITCH DECK

Presented by

**Claudia Alves**